

The Bernoulli Greedy-Energy Sequence from One Seed

Dyadic structure, logarithmic energy, and a quadratic-logarithmic obstruction

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Abstract

For the periodic second Bernoulli polynomial

$$B_2(x) = \{x\}^2 - \{x\} + \frac{1}{6},$$

consider a greedy sequence on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, started from one point, in which x_N minimizes $\sum_{k < N} B_2(x - x_k)$. Pausinger's theorem implies that these are permuted base-2 van der Corput sequences. A short kernel-specific argument gives more: every prefix is a disjoint union of regular dyadic grids whose sizes are exactly the nonzero binary digits of N . Consequently the potential is uniformly bounded by $1/3$, and the pair energy is at most $\text{pop}_2(N)/2 = O(\log N)$.

The same decomposition disproves the explicit weighted exponential-sum conjecture in Problem 23 of Steinerberger's list. For

$$T_N = \sum_{\ell=1}^N \frac{1}{\ell} \left| \sum_{k=0}^{N-1} e^{2\pi i \ell x_k} \right|,$$

one always has $T_N = O((\log N)^2)$, and for every admissible tie-breaking, $T_{(4^m-1)/3} \gtrsim m^2$. Thus the correct worst-subsequence order in the one-seed case is $\Theta((\log N)^2)$, not $O(\log N)$.

1 Statement of the result

All points and shifts below are understood modulo 1. For a power of two q , set

$$G_q(a) = \left\{ a + \frac{j}{q} : 0 \leq j < q \right\}, \quad M_q(a) = G_q\left(a + \frac{1}{2q}\right).$$

Thus $G_q(a)$ is a regular q -point grid and $M_q(a)$ is its midpoint coset. Let

$$P_N(x) = \sum_{k=0}^{N-1} B_2(x - x_k), \quad E_N = \sum_{k,l=0}^{N-1} B_2(x_k - x_l),$$

where x_0 is arbitrary and, for $N \geq 1$,

$$x_N \in \arg \min_{x \in \mathbb{T}} P_N(x).$$

No convention is imposed when the minimizer is not unique.

Theorem 1 (Exact one-seed structure and bounds). *For every greedy sequence above and every $N \geq 1$, write the binary expansion as*

$$N = q_1 + \cdots + q_s,$$

where the q_i are distinct powers of two. Then there are shifts $a_1, \dots, a_s$ such that

(i) *the prefix is a disjoint union*

$$\{x_0, \dots, x_{N-1}\} = \bigsqcup_{i=1}^s G_{q_i}(a_i),$$

and $\bigcap_{i=1}^s M_{q_i}(a_i) \neq \emptyset$;

(ii) *its potential satisfies the exact representation*

$$P_N(x) = \sum_{i=1}^s \frac{1}{q_i} B_2(q_i(x - a_i)), \quad \arg \min P_N = \bigcap_{i=1}^s M_{q_i}(a_i);$$

(iii) *uniformly in N and in all tie choices,*

$$\|P_N\|_{L^\infty(\mathbb{T})} \leq \frac{1}{3};$$

(iv) *with $s = \text{pop}_2(N)$, the pair energy obeys*

$$0 \leq E_N \leq \frac{s}{2} \leq \frac{1 + \log_2 N}{2}.$$

In particular, for one initial point the two qualitative questions in Problem 23 have stronger answers: the proposed logarithmic potential bound is actually constant, while the pair energy is logarithmic. The one-seed restriction matters; Pausinger explicitly leaves two or more arbitrary initial points as an open direction.

Theorem 2 (The proposed weighted exponential-sum bound fails). *For every one-seed greedy sequence, define*

$$A_N(\ell) = \sum_{k=0}^{N-1} e^{2\pi i \ell x_k}, \quad T_N = \sum_{\ell=1}^N \frac{|A_N(\ell)|}{\ell}.$$

Then

$$T_N \leq \sum_{i=1}^s H_{\lfloor N/q_i \rfloor} = O((\log N)^2),$$

where $H_r = \sum_{j=1}^r j^{-1}$. On the other hand, if

$$N_m = 1 + 4 + \dots + 4^{m-1} = \frac{4^m - 1}{3},$$

then, independently of every tie choice,

$$T_{N_m} \geq \frac{\log 2}{3} (m-1)^2.$$

More precisely, the proof gives

$$T_{N_m} \geq \frac{\log 2}{3} m^2 - O(m) = \frac{1}{12 \log 2} (\log N_m)^2 - O(\log N_m).$$

Therefore $T_{N_m} = \Theta((\log N_m)^2)$, contradicting the proposed bound $T_N \lesssim \log N$.

2 The dyadic decomposition

The only analytic input is the Bernoulli multiplication identity

$$\sum_{j=0}^{q-1} B_2 \left(x - a - \frac{j}{q} \right) = \frac{1}{q} B_2(q(x - a)). \quad (1)$$

It follows either from the Fourier series of B_2 or by checking the quadratic pieces. Since B_2 has its unique minimum $-1/12$ at $1/2$ modulo 1, the minimizer set of the left side of (1) is precisely $M_q(a)$.

Proof of Theorem 1(i)–(ii). We maintain the following invariant: the prefix is a disjoint union of grids of distinct dyadic sizes, those sizes are the nonzero binary digits of N , and their midpoint cosets have nonempty intersection. It is true at $N = 1$.

Assume it at N . By (1), each grid contribution is bounded below by $-1/(12q_i)$. The common intersection of the midpoint cosets attains all these lower bounds simultaneously. Hence the global minimizer set is exactly $\bigcap_i M_{q_i}(a_i)$.

Choose any x in this intersection. Let t be the number of trailing 1's in the binary expansion of N . Thus grids of sizes $1, 2, \dots, 2^{t-1}$ occur and a grid of size 2^t does not. Start with the new singleton $\{x\}$. It is the midpoint coset of the old singleton grid, so their union is a 2-grid. That 2-grid contains x and hence equals the midpoint coset of the old 2-grid; their union is a 4-grid. Continuing in this way performs the usual binary carry and produces one grid of size 2^t .

It remains to check the common-minimizer part of the invariant. If the new grid has size 2^t and contains x , then $z = x + 2^{-t-1}$ lies in its midpoint coset. Every untouched block has size $Q \geq 2^{t+1}$; its midpoint coset contains x and is invariant under translation by $1/Q$. Since 2^{-t-1} is an integer multiple of $1/Q$, it contains z as well. This completes the induction. Summing (1) over the blocks gives (ii). $\square$

Proof of Theorem 1(iii). On $\mathbb{T}$, $|B_2| \leq 1/6$. The q_i are distinct powers of two, so

$$\|P_N\|_\infty \leq \frac{1}{6} \sum_{i=1}^s \frac{1}{q_i} < \frac{1}{6} \sum_{r=0}^{\infty} 2^{-r} = \frac{1}{3}.$$

$\square$

3 Pair energy

The Fourier series

$$B_2(x) = \frac{1}{2\pi^2} \sum_{h \in \mathbb{Z} \setminus \{0\}} \frac{e^{2\pi i h x}}{h^2}$$

shows at once that

$$E_N = \frac{1}{\pi^2} \sum_{h=1}^{\infty} \frac{|A_N(h)|^2}{h^2} \geq 0. \quad (2)$$

For the upper bound, a grid has self-energy $1/6$. If $Q > q$ are dyadic powers, then

$$\begin{aligned} & \sum_{x \in G_Q(a)} \sum_{y \in G_q(b)} B_2(x - y) \\ &= \frac{q}{Q} B_2(Q(b - a)), \end{aligned}$$

because (1) first collapses the Q -grid and Q/q is an integer. Its absolute value is at most $q/(6Q)$. For each smaller block, the ratios to all larger distinct dyadic blocks sum to less than 1. Thus the total self-energy is $s/6$, the absolute value of the unordered cross-energy is less than $s/6$, and

$$E_N \leq \frac{s}{6} + 2\frac{s}{6} = \frac{s}{2}.$$

Together with (2), this proves Theorem 1(iv).

4 Weighted exponential sums

A q -grid has the exact exponential sum

$$\sum_{x \in G_q(a)} e^{2\pi i \ell x} = \begin{cases} qe^{2\pi i \ell a}, & q \mid \ell, \\ 0, & q \nmid \ell. \end{cases} \quad (3)$$

Therefore

$$|A_N(\ell)| \leq \sum_{q_i \mid \ell} q_i,$$

and summing in the opposite order gives

$$T_N \leq \sum_i \sum_{\substack{1 \leq \ell \leq N \\ q_i \mid \ell}} \frac{q_i}{\ell} = \sum_i H_{\lfloor N/q_i \rfloor} = O((\log N)^2).$$

For the lower bound, take $N = N_m = (4^m - 1)/3$. Its nonzero binary digits, and hence its block sizes in Theorem 1, are

$$1, 4, \dots, 4^{m-1}.$$

Fix $0 \leq j < m$ and let $\ell = 4^j u$ with u odd. By (3), precisely the blocks of sizes $1, 4, \dots, 4^j$ contribute. Whatever their phases, the largest one dominates all the smaller ones, and the reverse triangle inequality gives

$$|A_{N_m}(4^j u)| \geq 4^j - \sum_{r=0}^{j-1} 4^r = \frac{2 \cdot 4^j + 1}{3} \geq \frac{2}{3} 4^j. \quad (4)$$

Also

$$\left\lfloor \frac{N_m}{4^j} \right\rfloor = N_{m-j}.$$

If $H_R^{\text{odd}} = \sum_{1 \leq u \leq R, u \text{ odd}} u^{-1}$, summing (4) yields

$$T_{N_m} \geq \frac{2}{3} \sum_{r=1}^m H_{N_r}^{\text{odd}}. \quad (5)$$

Since $N_r \geq 4^{r-1}$ and

$$H_R^{\text{odd}} = \sum_{k=1}^{(R+1)/2} \frac{1}{2k-1} \geq \frac{1}{2} H_{(R+1)/2},$$

the terms with $r \geq 2$ give

$$H_{N_r}^{\text{odd}} \geq \left(r - \frac{3}{2}\right) \log 2.$$

Putting this into (5) proves

$$T_{N_m} \geq \frac{2}{3} \log 2 \sum_{r=2}^m \left(r - \frac{3}{2} \right) = \frac{\log 2}{3} (m-1)^2.$$

The sharper asymptotic form follows from $H_R^{\text{odd}} = \frac{1}{2} \log R + O(1)$.

5 Context and novelty boundary

The classification of one-seed greedy sequences as permuted van der Corput sequences is known from Pausinger [2]; the dyadic argument above is a short specialization to the Bernoulli kernel, included to make all estimates and all tie choices transparent. Brown and Steinerberger [1] posed logarithmic energy and potential bounds for this kernel and discussed the weighted exponential-sum criterion. The results above settle both energy questions for one seed, with a constant rather than logarithmic potential bound, and show that the stated weighted ℓ^1 criterion is too strong even in this most structured case. The arbitrary finite-seed problem is not addressed here.

References

- [1] L. Brown and S. Steinerberger, *Positive-definite functions, exponential sums and the greedy algorithm: a curious phenomenon*, J. Complexity **61** (2020), 101485. [doi:10.1016/j.jco.2020.101485](https://doi.org/10.1016/j.jco.2020.101485).
- [2] F. Pausinger, *Greedy energy minimization can count in binary: point charges and the van der Corput sequence*, Ann. Mat. Pura Appl. **200** (2021), 165–186. [doi:10.1007/s10231-020-00990-7](https://doi.org/10.1007/s10231-020-00990-7).